



CL 299 presentation 6

LNP Manufacturing and Process Simulation

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Energy

Syringe Pump :

Flow Rate	Power
1–10 mL/min	~10 W
10–50 mL/min	~15–20 W

Peristaltic Pump :

Flow Rate	Power
10–50 mL/min	20–40 W
50–100 mL/min	40–80 W

TFF circulation pump :- 80–120 W

Magnetic stirrer :- 5–15 W

Total power :- 240W

Process Simulation Code

- This code supports both forward and backward calculation.

Main components:

- `forward_calculation()`
 - `backward_calculation()` → Reverse calculation
 - `dof_analysis()` → System solvability
 - `solve()` → Mode selection
 - `run_simulation()` → Execution + output
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Forward Calculation

- Input Feeds: Organic and Aqueous Flow
 - $FRR = aq/org$
 - $Mixer = aq + org$
 - Dilution: $dilution = ratio \times mixer$
 $tff_in = mixer + dilution$
 - TFF: $retentate = tff_in / VRR$
 $permeate = tff_in - retentate$
 - Cryoprotectant Addition: $cryo_out = retentate + cryo_flow$
 - Final Filtration: $product = cryo_out - dead_volume$
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Backward Calculation

- **Goal:** Determine required inputs from desired product

Reverse steps:

- Add back filter loss, remove cryoprotectant
- Reverse TFF using VRR, reverse dilution
- Split mixer using FRR

Final result:

- Required aqueous flow
 - Required organic flow
 - Mixer conditions
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Simulation Workflow

- Function: `run_simulation()`

Steps:

- Read JSON input
 - Create run folder
 - Execute solver
 - Save outputs:
 - `output.txt`
 - `diagnostics.txt`
 - Zip all results
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PAT AS CONTROL SYSTEM

- pH sensor
 - flow meter
 - temp prob
 - conductivity sensor
 - Pressure sensor
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PAT NOT AS CONTROL SYSTEM

- DLS
 - Zeta potential analysis
 - Optical turbidity measurement
 - PVM
 - NIR
 - UV-Vis spectroscopy
 - Raman spectroscopy
 - HPLC
 - Charged aerosol detectors
 - RI
 - UV diode ray
 - LC-MS
 - NMR
 - Cryo-TEM
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UNIVERSAL PAT FOR LNP MANUFACTURING

- DLS
 - Zeta Potential analysis
 - pH sensor
 - flow meter
 - temp prob
 - conductivity sensor
 - Pressure sensor
 - Fluorescence-based assays
 - Cryo-TEM
 - Size Exclusion Chromatography
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Resources

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THANK YOU!!!